Prompt Verbosity as a Spectral Stabi

A Frequency-Filter Account of Vision–Language Model Robustness

Anonymous Submission

Verbose prompts make VLMs more robust — and we explain why.

$$\boxed{S_{\mathrm{pred}}(t, p) \;=\; \sum_{\omega} |W_t(\omega)|^2 \, |\Delta F(\omega)|^2}$$

Motivation

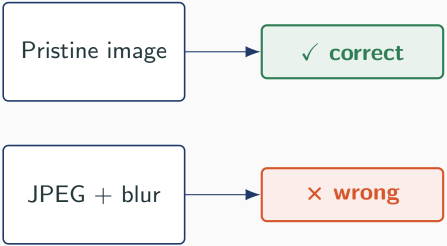
Why VLM Robustness Matters Now

Where VLMs are already deployed:

- Clinical triage & radiology pre-screening
- Assistive scene description for blind users
- Invoice / driver-licence / document OCR
- Autonomous perception and embodied agents

What the input actually looks like:

- Phone snapshot, wrong angle
- JPEG / motion blur / low light
- Sticker, shadow, partial occlusion



same image, same model

A deployment team has no principled way to ask:
"Will this model be safe on *this task* under *this corruption profile*?"

The Gap in the Literature

Benchmarks
VQA-CP, Winoground, ARO, CREPE
↓
black-box, no prediction

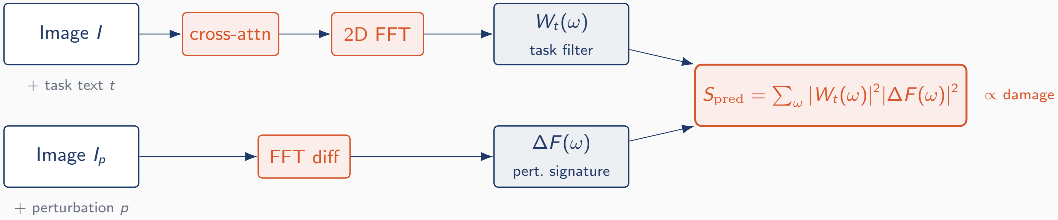
Fourier vision
Yin, Wang, Tsuzuku
↓
task-free, ignores text

Probing
cross-modal attention
↓
describes, not predicts

No framework predicts how much a VLM will break before the perturbation is applied.

The Core Idea

Our Thesis in One Picture



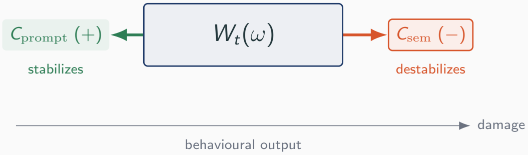
Text is a physical operator on the image's Fourier spectrum.
Damage follows the *overlap* S_{pred}, not the raw magnitude $\|\Delta F(\omega)\|$.

Can we control the damage?

Semantic program complexity C_{sem}
reasoning logic — entities, attributes, relations, operators, depth, grounding ambiguity.

Prompt load C_{prompt}
lexical surface — content-token count after stopword removal.

Natural questions confound them.
Raw “complexity” slopes in prior work mix two effects whose signs *oppose each other*.



The mirror 8-level design separates the two at the item level.

Contributions at a Glance

Headline

1. Verbose prompts stabilise VLMs. 65–86% drift-variance reduction across mirror pairs (mean ratio 0.247); mean volatility drops 68–90% on 8B, strongest at L4. Not previously reported.

2. Spectral–behavioural law $S_{\text{pred}} = \langle W_t(\omega)^{(\texttt{last\_2})}, \Delta F(\omega) \rangle$ *explains* the stabilisation at $r = 0.337$ grouped, $r = 0.315$ per-sample, and within-cell median $r = 0.543$ (2D, question-only). Zero target-side fitting.

3. Four-stage layer trajectory: layer-0 length prior $\rightarrow$ erasure $\rightarrow$ mid-layer content narrowing $\rightarrow$ late + `last_2` verbosity-stabilising gap.

Method & Evidence

4. Mirror 8-level GQA design (L1–L8) isolates C_{sem} from C_{prompt} at the item level. Within-image FE recovers the tug-of-war: $\beta_{C_{\text{prompt}}} < 0$, $\beta_{C_{\text{res}}} > 0$.

5. Program-aware complexity metric grounded in GQA/CLEVR, NMN, referring expressions, psychometrics.

6. Reproducible, deployable pipeline — released as a modular repository. Prompt padding is the cheapest training-free defence.

Theoretical Framework

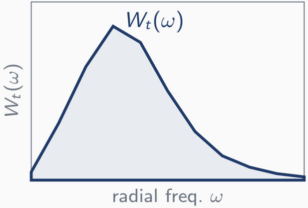
The Task-Specific Filter $W_t(\omega)$

1. Vision tokens $\{v_{ij}\}$ on an $H \times W$ grid.
2. Cross-modal attention at layer group ℓ gives spatial map $a_t^{(\ell)}$.
3. 2D FFT, square, radially bin into B linear bands.
4. Average over images; L1-normalise:

$$W_t(\omega)^{(\ell)} = \frac{\mathbb{E}_I[\,|\mathcal{F}_2\, a_t^{(\ell)}(I)|^2(\omega)\,]}{\sum_{\omega'} \mathbb{E}_I[\ldots]}$$

Effective bandwidth via IPR (L1-normalised probability):

$$G(t) = \left(\sum_{\omega} W_t(\omega)^2\right)^{-1}, \quad 1 \le G(t) \le B$$



Illustrative task-conditioned radial filter.

The Overlap Law

Perturbation spectral signature:

$$\Delta F(\omega)_p(\omega) = \mathbb{E}_I\left[\,\left|\,|\mathcal{F}_2 I_p|^2 - |\mathcal{F}_2 I|^2\,\right|\,\right]_\omega$$

Model-independent; purely a property of p.

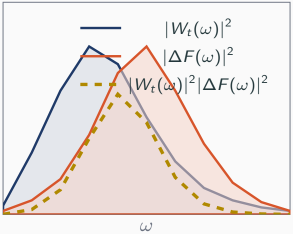
Two forms, both parameter-free:

$$S_{\text{pred}}^{\text{quad}} = \textstyle\sum_\omega |W_t(\omega)|^2 |\Delta F(\omega)|^2$$
$$S_{\text{pred}}^{\text{lin}} = \langle W_t(\omega), \Delta F(\omega)\rangle = \textstyle\sum_\omega W_t(\omega) \cdot \Delta F(\omega)$$

Central claim:

$$\mathbb{E}[\,\Delta\text{acc}(t,p)\,] \;\propto\; S_{\text{pred}}(t,p)$$

Damage follows the overlap, not raw energy.



S_{pred} is the area under the gold curve.

Semantic Program Complexity C_{sem}

$$C_{\mathrm{sem}} = E + A + R + O + D + \ln(1 + G_{\mathrm{amb}})$$

- E — entity refs (GQA `select`/`query`)
- A — attribute refs (GQA `query_attr`)
- R — relation refs (GQA / CLEVR `relate`)
- O — reasoning ops (NMN primitives)
- D — program depth (CLEVR)
- G_{amb} — grounding ambiguity (ReferIt)

Equal weighting preserves L1 $<$ L2 $\lesssim$ L3 $<$ L4 under *any* positive linear combination
$\Rightarrow$ the sign of every effect is invariant to the weight scheme.

Regression Specification

Residualise, then horse-race (primary target: loglik volatility)

$$C_{\mathrm{res}} = C_{\mathrm{sem}} - \hat{C_{\mathrm{sem}}}(C_{\mathrm{prompt}})$$

$$y_{i,t} = \alpha_i + \beta_{C_{\mathrm{sem}}} C_{\mathrm{sem}}(t) + \beta_{C_{\mathrm{prompt}}} C_{\mathrm{prompt}}(t) + \beta_H H_{\mathrm{opt}}(t) + \varepsilon_{i,t}$$

Primary

Log-likelihood volatility

$\mathbb{E}_p\big[\,|\log p_\theta(a^\star \mid I_p, t) - \log p_\theta(a^\star \mid I, t)|\,\big]$

mean absolute drift (L1)

Secondary

Gated accuracy drop

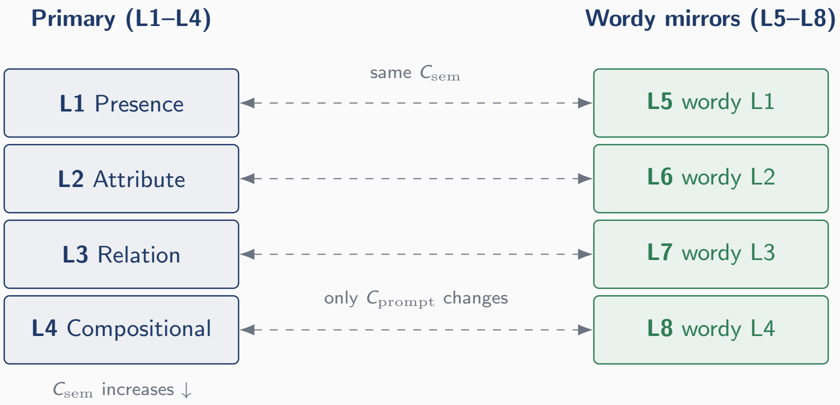
clean-correct $\rightarrow$ perturbed-wrong

thresholded, binary, confirmatory

Paired mirror t-tests (L1↔L5, L2↔L6, L3↔L7, L4↔L8) identify the C_{prompt} effect without regression assumptions.

The Mirror 8-Level Design

The Mirror 8-Level GQA Design



L5–L8 preserve the semantic atoms byte-for-byte of their L1–L4 counterparts. Only prompt load C_{prompt} changes. Yes/no levels balanced 50/50 by complementary pattern pairing.

Perturbation Bank and Scale

14-class perturbation bank:

- **Geometric** — translation, pad-crop, scale, scale-and-pad $\times 2$, rotation (6)
- **Occlusion** — text overlay, box overlay, random-text (3)
- **Spectral** — lowpass, highpass, low-/high-/all-band noise (5)

Severities $\{1, 2, 3\}$.

Model & evaluation scale

Qwen3-VL-2B-Instruct

Attention extracted at *all* decoder layers

Partitioned into early / mid / late thirds

1000 GQA images

8000 (image, level) points

Full 8-level mirror grid

$B = 20$ radial bands (auto-selected)

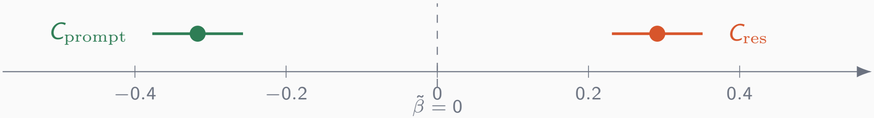
No Hann taper, no DC suppression

Results

Result 1a: The Tug-of-War on Volatility

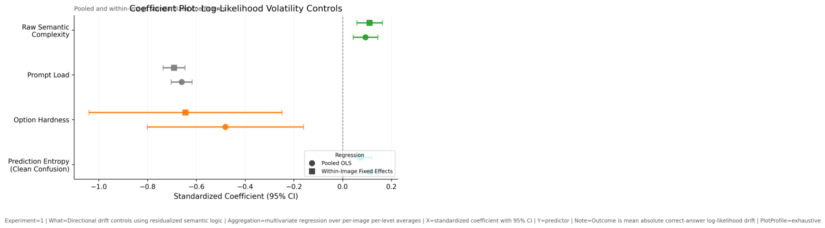
Log-likelihood volatility $\mathbb{E}_p\big[\,\lvert \log p_\theta(a^\star \mid I_p, t) - \log p_\theta(a^\star \mid I, t)\rvert\,\big]$, $n = 8000$:

Predictor	**Pooled OLS $\tilde\beta$**	**Within-image FE $\tilde\beta$**
paccent!8C_{res} (residualised C_{sem})	paccent!8$+0.266$ $(p<10^{-107})$	paccent!8$+\mathbf{0.291}$ $(\mathbf{p<10^{-130}})$
pgreen!8C_{prompt}	pgreen!8-0.296 $(p<10^{-177})$	pgreen!8$-\mathbf{0.317}$ $(\mathbf{p<10^{-209}})$
H_{opt} (control)	$+0.090$ $(p<10^{-13})$	$+0.092$ $(p<10^{-14})$
n / R^2	8000 / 0.187	8000 / **0.217**



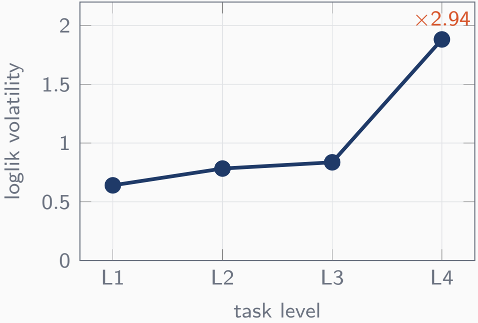
Opposite signs, jointly significant at p in the 10^{-100}s.

Result 1a: Forest Plot of the Tug-of-War



Standardised regression on the primary target (loglik volatility): C_{res} destabilises (+, top), C_{prompt} stabilises (−, middle); pooled OLS and within-image FE agree. 100-sample 8B replication.

Result 1b: Strict Monotonicity in Volatility



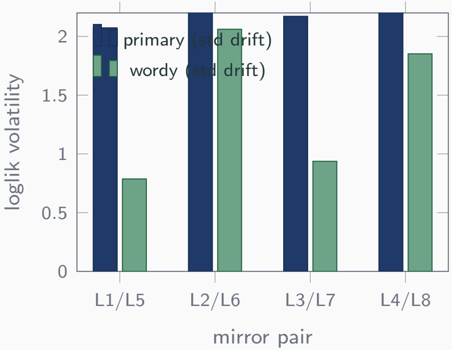
Strictly monotone L1 $\to$ L4:

- L1 $= 0.640$
- L2 $= 0.783$
- L3 $= 0.836$
- L4 $=$ **1.882**

Kendall $\tau = 1.0$
Cohen's $d_{\text{L4 vs L1}} = 1.43$
ANOVA $F = 255$, $p < 10^{-139}$

The spectral cost of reasoning — in one curve.

Result 1c: The Mirror Contrast



Same semantic program; only C_{prompt} changes. Bars: drift std deviation; ladder: drift-variance reduction.

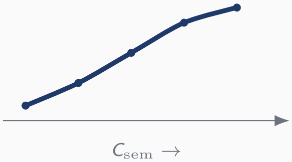
L1↔L5	−**86%**
L2↔L6	−64%
L3↔L7	−81%
L4↔L8	−70%

Every wordy mirror stabilises its primary.
The anchor is largest where reasoning is costliest.

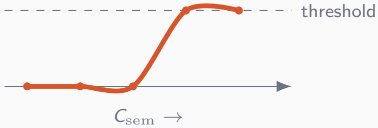
Result 1d: Why Volatility, Not Accuracy

Quantity	Primary: volatility	Secondary: accuracy drop
pnavy!6FE R^2	pnavy!6**0.217**	0.020
FE $\tilde{\beta}(C_{\mathrm{res}})$	$+0.291$	$+0.172$
FE $\tilde{\beta}(C_{\mathrm{prompt}})$	-0.317	-0.033^*
pnavy!6Monotonicity L1→L4	pnavy!6Kendall $\tau = 1.0$	Kendall $\tau = 0.67$ (one reversal)

Continuous (volatility)



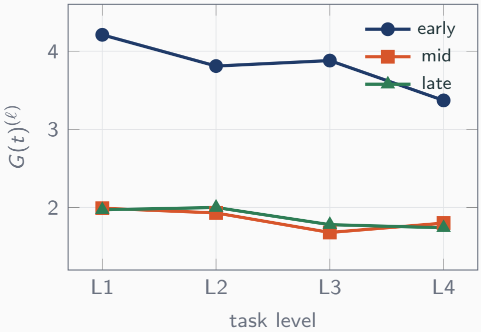
Thresholded (accuracy)



Accuracy is a lossy $\{0,1\}$ projection. Shifts that don't cross the boundary are invisible.

Result 2: Layer-Resolved Bandwidth

Effective bandwidth $G(t)^{(\ell)}$ across L1–L4:



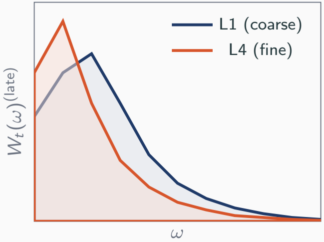
- Spearman $\rho_{\text{overall}} = -1.0$
- Each layer group $\rho = -0.80$
- ANOVA: $F = 255$, $p < 10^{-139}$

Early layers: wider (≈ 3.8), track C_{prompt}.

Late layers: concentrated (≈ 1.9), track C_{sem} alone.

Early C_{prompt}-anchoring gives way to late C_{sem}-convergence.

Late-Layer Downward Concentration



Why does $G(t)^{(\text{late})}$ fall with C_{sem}?

Mass concentrates *downward* onto the lowest one or two bins:

- top-2 mass **rises** monotonically with C_{sem}.
- high-freq tail mass **falls** monotonically.
- $\rho(G(t)^{(\text{late})}, \text{rank}) = -1.0.$

Mass does **not** climb upward.
It piles up onto DC & DC-adjacent bins — exactly where natural perturbation $\Delta F(\omega)$ also lives.

Shape Signatures (Late Layer)

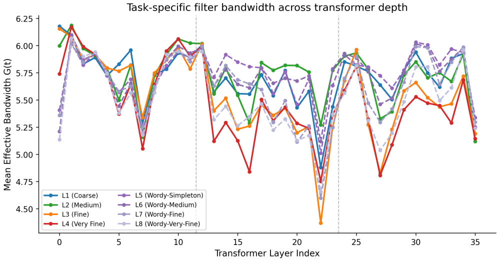
Because $G(t)$ conflates spread and location, we report direct shape statistics on the late-layer filter:

Statistic	**L1 (coarse)**	**L4 (fine)**
top-2 mass (two largest bins)	lower	**higher** ($\rho > 0$)
high-freq tail mass	larger	**thinner** ($\rho < 0$)
top-3 mass (three largest bins)	—	informational
centroid along ω	—	shifts slightly down

Downward concentration: the late-layer filter for finer tasks pools its mass into the lowest one or two bins. The high-frequency tail *thins* rather than fattens. Earlier "upward migration / tail growth" framing is retracted on these data.

Top-2 mass and tail mass are the two load-bearing gates; both flip sign vs the earlier prediction. $G(t)$ alone misses the location story.

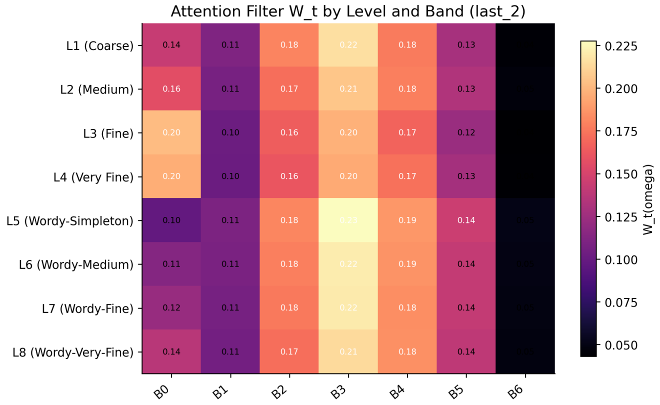
Empirical Four-Stage Trajectory (8B, 100 samples)



Experiment=2 | What=Per-layer effective bandwidth without early/mid/late binning | Aggregation=sample average by level and decoder layer | X=decoder layer index | Y=mean effective bandwidth G(t) | Note=Solid=L1-L4 primary; dashed=L5-L8 wordy; vertical dashed lines mark layer-group boundaries

Layer 0 length prior (-0.74 gap) $\rightarrow$ erasure (layers 1–12) $\rightarrow$ mid-layer content narrowing $\rightarrow$ `last_2` verbosity gap ($+0.33$ peak at layer 31).

Filter Shape Across the 8 Levels (`last_2`)



Experiment=2 | What=Layer-group task filter W_t (last_2) | Aggregation=level-wise sample average | X=frequency band (low→high) | Y=task level | PlotProfile=exhaustive

Primary L1–L4 (top) concentrate mass downward as C_{sem} rises; wordy L5–L8 (bottom) keep a broader, more uniform mass distribution at the same depth.

Theorem 4 — Four-Stage Trajectory & Length Prior

Claim. $W_t(\omega)(\ell)$ moves through *four* stages with depth (Qwen3-VL-8B, 36 decoder attention layers):

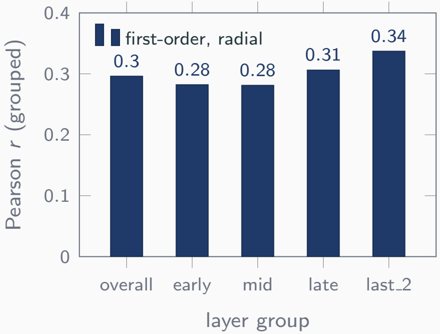
- **Layer 0:** prompt-length prior — wordy mirrors -0.74 narrower than primaries in mean bandwidth (largest single-layer gap anywhere in network).
- **Layers 1–12:** prior erased — gap collapses by layer 2; all 8 levels equilibrate near bandwidth ≈ 6.0, length and content blind.
- **Layers 13–23 (mid):** C_{sem}-driven narrowing, length-insensitive — L4 dips to $G(t) = 4.4$ at layer 22; primary $\approx$ wordy throughout.
- **Layers 24–35 + `last_2`:** content narrowing + verbosity gap — wordy filter *broader* than primary, peaking at $+0.33$ at layer 31.

Why $\beta_{C_{\mathrm{prompt}}} < 0$? The deep-decoder verbosity gap lands the wordy filter on a broader, less DC-exposed shape, lowering $\langle W_t(\omega)^{(\mathtt{last\_2})}, \Delta F(\omega)\rangle$ linearly. Drift drops mechanically.

Wordy prompts trigger a sharp layer-0 length prior, get equalised at early layers, undergo content narrowing without surface-form bias at mid, then open a stabilising gap at late and `last_2`.

Result 3: Overlap Predicts Behavioural Drift, Layer by Layer

Pixel-space first-order overlap $\langle W_t(\omega)^{(\ell)}, \Delta F(\omega)\rangle$ vs $\mathrm{loglik\_volatility}$ (100-sample 8B run, $B{=}7$):

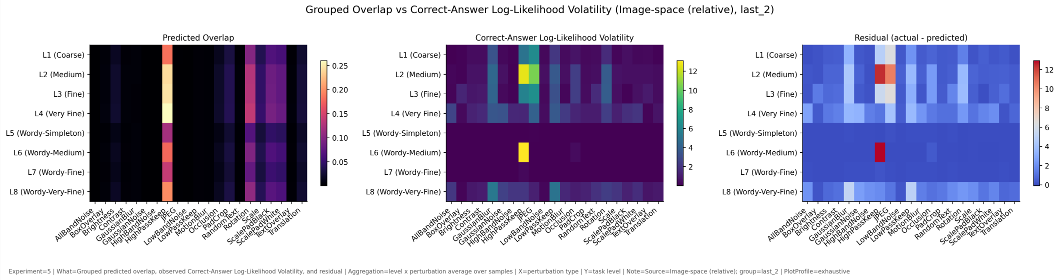


All cells significant ($p \approx 0$, $n = 16{,}800$).

- **last_2** carries the strongest signal: $r = 0.337$ grouped, $r = 0.315$ per-sample.
- Per-level $r \in [0.22, 0.38]$.
- 2D first-order at `last_2`: $r = 0.315$ grouped.
- Quadratic variant $r = 0.14$ (legacy); first-order wins by $\approx 2.4\times$.

> $W_t(\omega)^{(\mathrm{last\_2})}$ carries the strongest spectral coupling to perturbation damage — the second-from-last attention layer is where the task-conditioned filter is sharpest.

Overlap Law: Predicted vs Observed vs Residual (8B, `last_2`)



Predicted overlap (left) and observed volatility (middle) light up the same (level, perturbation) cells; residual (right) is mostly noise. $r = 0.337$ grouped, 0.315 per-sample, zero target-side fitting.

Matched-FE: Item-Level Law + Vision-Feature Sign Flip

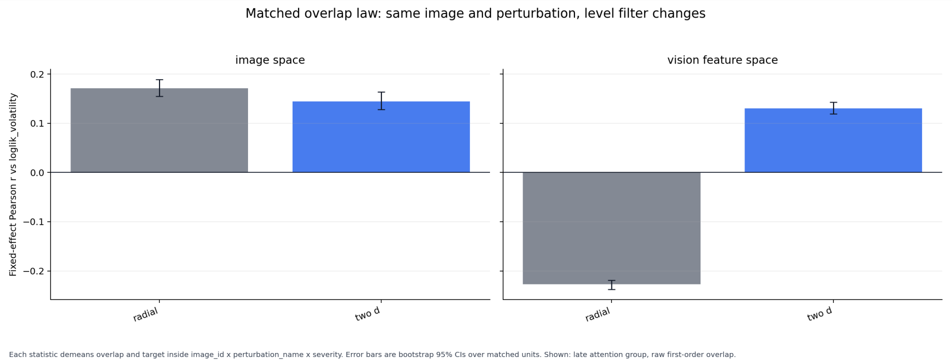


Image-space: both radial and 2D positive (law works within-cell). **Feature-space:** radial flips negative; 2D recovers positive — the apparent "sign flip" is a radial-reduction artefact. Within-cell median Pearson at `last_2`: 0.350 radial / 0.457 2D / **0.543** 2D + question-only.

Results 4 & 5: Cutoff Law and Overlap Validation

Task 4 — Frequency cutoff $\omega_c(C_{\text{sem}})$

- Largest ω_c at which gated accuracy drops $< \delta = 0.02$.
- Falls monotonically L1 $\to$ L4.
- Wordy mirrors inherit the cutoff of their primary.

Fine reasoning lives in the high-frequency tail.

Task 5 — Overlap law $\langle W_t(\omega)^{(\text{last\_2})}, \Delta F(\omega) \rangle$

- Grouped: $r = 0.337$ pooled radial, $r = 0.315$ 2D ($n = 16{,}800$).
- Per-sample: $r = 0.315$ ($n = 62{,}400$).
- Matched-FE within-cell median Pearson: 0.350 radial, 0.457 2D, **0.543** 2D + question-only.
- Zero target-side fitting.

The law works at both population and item level.

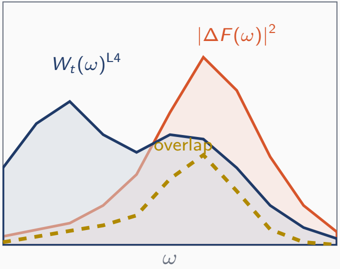
Grouped $r = 0.34$ shows the population law; within-cell $r = 0.54$ shows the item-level signal once orientation is preserved (2D) and answer-format tokens stripped from

Discussion & Impact

Why Reasoning Is Spectrally Costly

As C_{sem} rises (deep-decoder filter):

1. $W_t(\omega)^{(\mathtt{last\_2})}$ mass concentrates **downward** onto the top-1/top-2 bins (DC & near-DC).
2. High-freq tail **thins**.
3. Natural perturbations $\Delta F(\omega)$ also live on DC & near-DC (80–96% of mass).
4. $\langle W_t(\omega)^{(\mathtt{last\_2})}, \Delta F(\omega)\rangle$ grows $\Rightarrow$ damage follows.



Fine reasoning is spectrally costly because the second-from-last-layer filter concentrates onto exactly the bins where natural perturbations also live.

Why Length Anchors: The Four-Stage Trajectory

Layer 0
length prior
wordy -0.74
narrower

→

Layers 1–12
prior erased
all levels ≈ 6.0

→

Mid (13–23)
C_{sem}-narrowing
length-insensitive

→

Late + `last_2`
content + verbosity
wordy $+0.33$ broader
(layer 31 peak)

Length anchors because the layer-0 length prior + deep-decoder verbosity gap land the wordy filter on a broader, less DC-exposed shape than its primary counterpart—reducing $\langle W_t(\omega)^{(\mathtt{last\_2})}, \Delta F(\omega)\rangle$ and the drift it predicts.

Reconciling Prior Results

Positive complexity slope

Winoground, ARO, CREPE

Negative complexity slope

Sclar, Mizrahi (prompt sensitivity)

Both are right.

They measure the *sum* of two opposite-signed forces in corpora with different mixes. Residualising C_{sem} against C_{prompt} separates them cleanly.

Societal Impact: Silent Failure → Auditable Quantity

Radiology pre-screen

Estimate $W_t(\omega)$ on representative scans, intersect with corruption spectra (motion blur, JPEG, anonymisation) — read off fragility without a full behavioural sweep per model update.

Assistive scene desc.

Detect in advance that fine attribute queries on low-light photographs are more brittle than coarse presence queries. Adjust prompt or confidence accordingly.

Autonomous audits

A parameter-free number estimated from the model's own attention — not a held-out benchmark that may not resemble deployment.

Fairness dividend. Users who phrase questions in longer, less direct language (non-native speakers, assistive dictation, clinical / legal domains) sit on the *more robust* side of the tug-of-war. A finding invisible to any single-axis "prompt complexity" treatment.

Practical, Training-Free Interventions

1. Prompt padding

Deliberately verbose prompts hold early-layer mass in the low-freq floor.

A cheap, inference-time defence against high-freq corruption. Costs **zero** at train time.

2. Targeted augmentation

Augment with perturbations whose $\Delta F(\omega)$ *overlaps* the fine-task $W_t(\omega)$.

Not ImageNet-C in bulk — the specific bands the deployment task reaches into.

3. Spectral budget

Before deployment, compute $\omega_c(t)$ for the task of interest.

Sets a principled ceiling on how much low-pass compression pre-processing tolerates.

A Falsifiable Prediction

If the C_{prompt} anchor is genuinely early-layer in origin, then

ablating or perturbing early-layer cross-modal attention should preferentially dissolve the C_{prompt} effect while leaving the C_{sem} effect intact.

This is a **falsifiable mechanistic prediction** that purely behavioural accounts of prompt sensitivity cannot make.

Natural follow-up tools:

- Activation-patching (**?**)
- Causal mediation analysis
- Cross-model replication (LLaVA-NeXT, InternVL)

Limitations & Conclusion

Limitations

What's not yet validated

- Two Qwen3-VL scales (2B for 1000-image regression; 8B for 100-sample replication); LLaVA-NeXT / InternVL still ongoing.
- Severity-1 only in the main run.
- Task 4 (cutoff) reported qualitatively.

What's inherent

- $W_t(\omega)$ from attention $\Rightarrow$ correlational lower bound.
- Pixel-space radial Pearson–Spearman gap: $r = 0.34$ vs $\rho = 0.19$ at `last_2`. The 2D form and vision-feature space close this gap.
- Radial wins between-cells ($r = 0.34$); 2D wins within-cells ($r = 0.46$ median). Unified estimator open work.
- C_{sem} equal-weighted by design; signs invariant under positive reweighting, magnitudes are not.
- Mirror controls are synthetic; human-judged paraphrases are a natural extension.

Conclusion

Verbose prompts make VLMs measurably more robust to image perturbations — and we explain why.

What we found.

- **Headline:** every wordy mirror L5–L8 has 65–86% lower drift variance than L1–L4 (mean ratio 0.247 on 8B, replicating 2B's 0.246); mean log-likelihood volatility drops 68–90% on 8B.
- **Mechanism:** text is a **spectral operator** $W_t(\omega)$; verbose prompts *broaden* $W_t(\omega)^{(\mathtt{last\_2})}$ at the deepest decoder layer, shrinking overlap with natural-perturbation $\Delta F(\omega)$.
- **Predictor (population):** $\langle W_t(\omega)^{(\mathtt{last\_2})}, \Delta F(\omega) \rangle$ predicts damage at $r = 0.337$ pooled grouped, $r = 0.315$ per-sample, zero target-side fitting.
- **Predictor (item-level):** matched-FE within-cell median Pearson $r = 0.543$ (2D, question-only $W_t(\omega)$). The law works *within-cell*, not just between-cell.
- **Trajectory:** layer-0 length prior → erasure (1–12) → mid-layer content narrowing → late

Thank You.

Questions?

Backup: C_{sem} Axes

$$C_{\mathrm{sem}} = E + A + R + O + D + \ln(1 + G_{\mathrm{amb}})$$

- E — entity refs (GQA `select`/`query`)
- A — attribute refs (GQA `query_attr`, CLEVR)
- R — relation refs (GQA / CLEVR `relate`)
- O — reasoning ops (NMN primitives)
- D — program depth (CLEVR's key difficulty axis)
- G_{amb} — grounding ambiguity (log form, ReferIt)

Backup: ΔZ — Mean L2 Drift

$$\Delta Z(t,p)^{(\ell)} = \frac{1}{N^{(\ell)}} \sum_{n=1}^{N^{(\ell)}} \left\| z_n^{\mathrm{pert},(\ell)} - z_n^{\mathrm{clean},(\ell)} \right\|_2.$$

- Mean ℓ_2 over full multimodal sequence.
- Language tokens aligned positionally.
- Vision-token spans aligned with patch-grid-aware resampling.

Backup: All Confidence-Space Targets

Target (within-image FE, $n = 8000$)	R^2	$\tilde{\beta}(C_{\mathrm{res}})$	$\tilde{\beta}(C_{\mathrm{prompt}})$	$\tilde{\beta}(H_{\mathrm{opt}})$
pnavy!8**Loglik volatility (primary)**	**0.217**	+0.291	−0.317	+0.092
Loglik erosion	0.185	+0.308	−0.284	+0.032
Loglik recovery	0.140	−0.176	+0.281	−0.118
Loglik drift	0.016	+0.126	−0.056	−0.030
Accuracy drop (confirmatory)	0.020	+0.172	−0.033	−0.103

Volatility wins on R^2, on the C_{prompt} effect, and on both p-values.